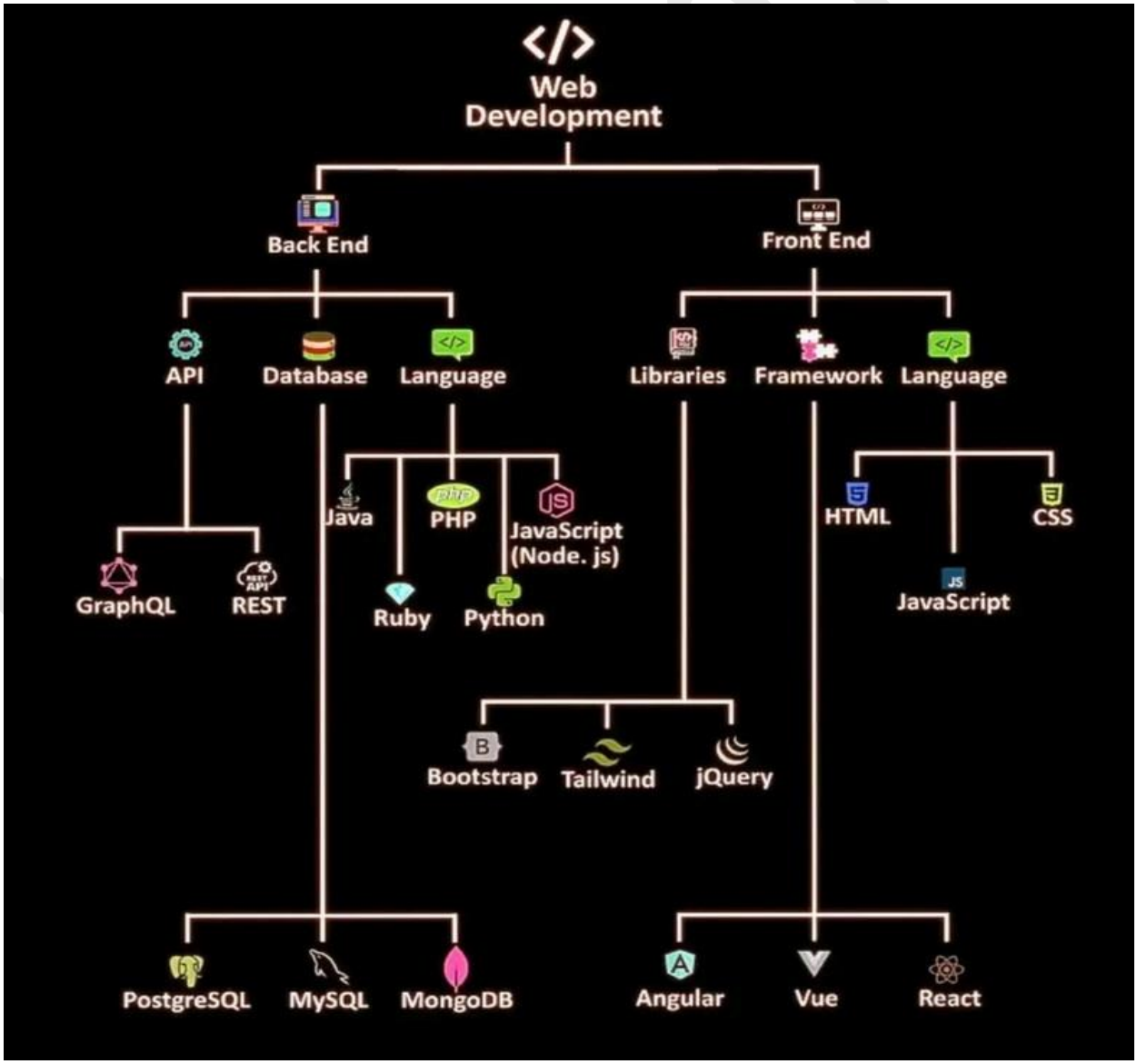
S.P.M College,Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1) **– Hira Kumar**

Web Development



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❖ Web / WWW

- The term “Web” Stands for the “World Wide Web(WWW)”.
- The Web is a vast information system on the internet where documents (web page) and other resources are interconnected by hyperlinks and URL’s and can be accessed through a web browser. (web, internet पर मौजूद जानकारी का एक विशाल जाल है, जिसमें सभी वेबसाइट और web page आपस में जुड़े होते हैं |)

(सीधी शब्दों में - internet एक सड़क की तरह हैं और web (WWW) उस सड़क पर चलने वाली कार है, जिसमें आप बैठकर दुनिया की किसी भी जानकारी(website) तक पहुँच सकते हैं)

❖ Types of Web

There are mainly 3 types of Web:

1. **Static Web**
2. **Dynamic Web**
3. **Semantic Web**

1. Static Web

- A static web page shows the same content to every user.
- e.g. – newswebsite, blogger page, Simple informational websites,…….
- It is created using HTML and CSS.

2. Dynamic Web

- A dynamic web page changes its content automatically.
- It uses programming languages and databases.
- Uses server-side languages like PHP, Python, JavaScript
- Example: Facebook, Amazon, YouTube,online form filling,.

3. Semantic Web

- The Semantic Web helps computers understand the meaning of data on the web.

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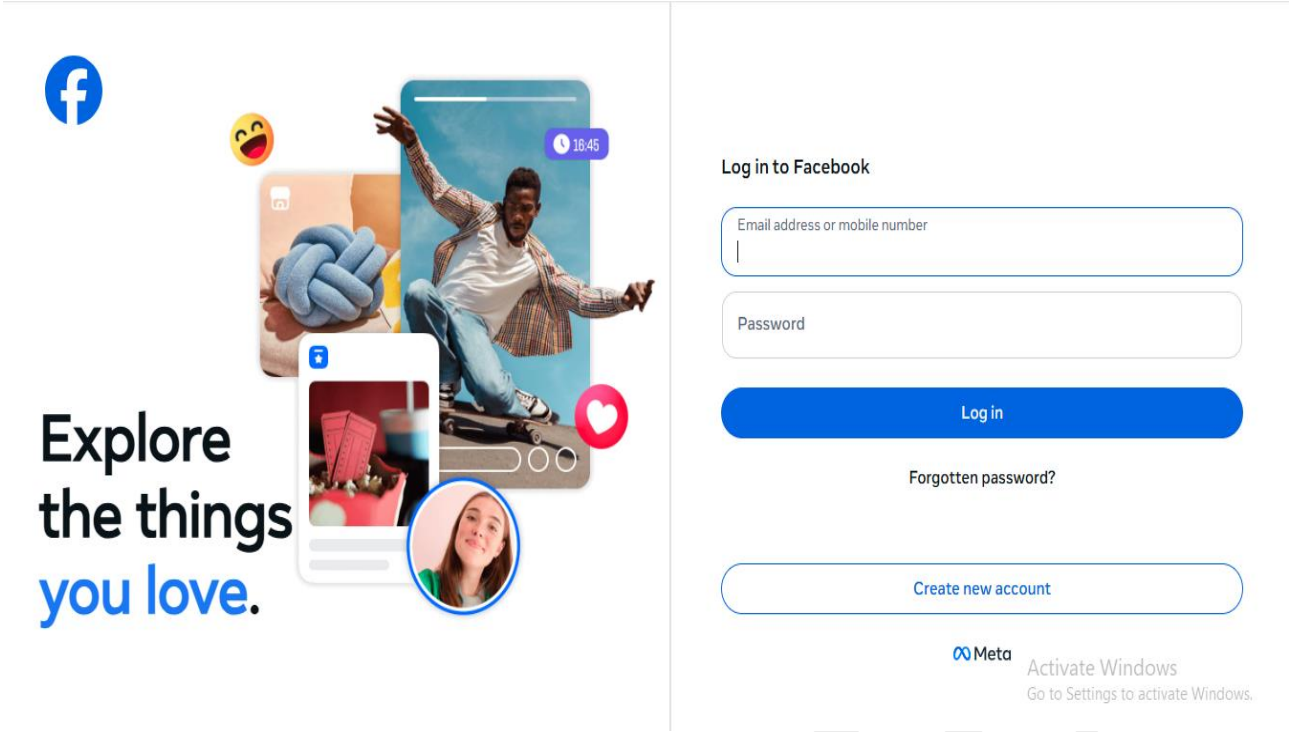
- It organizes information in a way that machines can process easily.
- Example - Smart search results and AI-based web services,

❖ Web page

- A web page is a single document available on the world wide web(WWW) that can be viewed using a web browser.
- Web pages are written using HTML and are stored on web servers.

What contain in web page ?

- In web page can contain text, image,video,link, etc…
- E.g. → Home page of google, Login page of facebook, contact page, about page,……



❖ What is website ?

- A Website is a collection of related web pages stored on a web server and accessed through the internet using a single domain name.

OR

- A website is a group of interconnected web pages that are accessed through a common domain name on the internet.

OR

- Collection of Web page is called Website.
- Example :- Wikipedia.org, ndtv.in, bbc.com, indianrailways.gov.in, imdb.com, pmkisan.gov.in, https://hira24y7.github.io/bcanotes/ , amazon.com, facebook.com, Zomato,….

❖ Features of Website

- ✓ Contains many web pages (e.g. – home page, product page, cart page, login page, contact page, about page,……)
- ✓ Hosted on a web server
- ✓ Accessed using a domain name
- ✓ Viewed through a web browser
- ✓ Includes text, images, videos and links

Types of Website

Two types

(i) Static Website

(ii) Dynamic Website

(i) Static Website

- A static websites is a website whose content remains the same for every user. IT is created using HTML and CSS and does not change automatically.
- Example – Personal portfolio, news website, Wikipedia,..

(ii) Dynamic website

- A dyamic website is a website whose content can change according to the user or data.
- It uses server-side language like PHP,Python, JS and a database.
- Example – online form filling, E-commerce website,social media websites.

Note – **Interactive website** → which website in which interaction b/w user and website.

Example – comment, button click, chat,…..

❖ What is Web Application ?

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- A web application (web app) is a software program that runs on a web server and is accessed by the user through a web browser (like chrome, safar, firefox,…) over the internet.
- Unlike traditional desktop software, you don't need to install a web application on your computer; you simply use it by visiting a website.

(web application एक ऐसा software प्रोग्राम हैं जिसे आप अपने कंप्यूटर या मोबाइल पर install किए बिना, सीधे internet browser के जरिए इस्तेमाल करते हैं)

- E.g. – amazon.com, facebook.com, irctc.com, makemytrip.com, bookmyshow.com, Zomato,….

❖ Difference b/w Website & Web application

Website	Web application
✓ Primarily designed to provide information to the visitor / user. ✓ It is mostly read only. ✓ All website are not web application. ✓ E.g. – A news site, a blog,…	✓ Designed to allow the user to perform tasks, manipulate data, and interact with the system. ✓ It can be read, write, and process. ✓ All web applications are websites, but not all websites are web applications. ✓ E.g. – Gmail, Facebook, Online Banking, Online Shoping,…..

❖ Web server

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- A web server is a system that stores website files and delivers web pages to users over the internet when requested by a browser.
- Example → Apache server, nginx server, Microsoft IIS, Google web server,…..

❖ What is Web Developer ?

- A Web developer is a professional who specializes in building and maintaining websites and web applications.
- They are the architects and builders behind the sites we use every day.
- Their role involves not just creating the visual look of a site but also ensuring it functions correctly, loads quickly, and provides a seamless user experience.

(web डेवलपर वह व्यक्ति हैं जो वेबसाइटों और web application को बनाने और maintain रखने का काम करता हैं | वे internet पर मौजूद उन सभी साईटों की रीढ़ होती हैं जिन्हें हम रोजाना इस्तेमाल करते हैं | उनका काम सिर्फ एक खुबसूरत दिखने वाली वेबसाइट बनाना ही नहीं, बल्कि यह देखना होता हैं की वह तेजी से लोड हो एवं सही तरीके से काम करें और उपयोग एवं user friendly हो |)

✓ **One line answer** → They build and maintain websites.

- Work :- Create Website, Web application, design + coding + maintain database,…
- E.g. → You (आप भी एक web developer हो सकते हैं)

❖ What is Web Development ?

- Web development is the work / process involved in building and maintaining websites and web applications. (web development - वेबसाइट या web application बनाने की प्रक्रिया हैं)
- It’s the process of creating everything you see and interact with on the internet.

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❖ **Web developers are typically categorized into three main types :**

1. **Front - End Developer**
2. **Back - End Developer**
3. **Full - stack Developer**

1. Front – End Developer

- They focus on the client-side or the part of the website that users see and interact with directly.
- This includes the layout, design, buttons, menus, and overall look and feel.
- They use languages like – HTML, CSS, and JavaScript.

2. Back – End developer

- They work on the server – side, which is the behind the scenes part of a website.
- They are responsible for the server, databases, and application logic that power the website.
- They ensure that the front -end has something to work with.
- They use languages and tools include – PHP, Python, Java, Node.js and SQL database.

3. Full – Stack developer

- They are proficient in both front-end and back-end development.
- A full-stack developer has the skills to handle all aspects of building a website or application, form the user interface to the server infrastructure.

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❖ **Web application are typically categorized into two main types :**

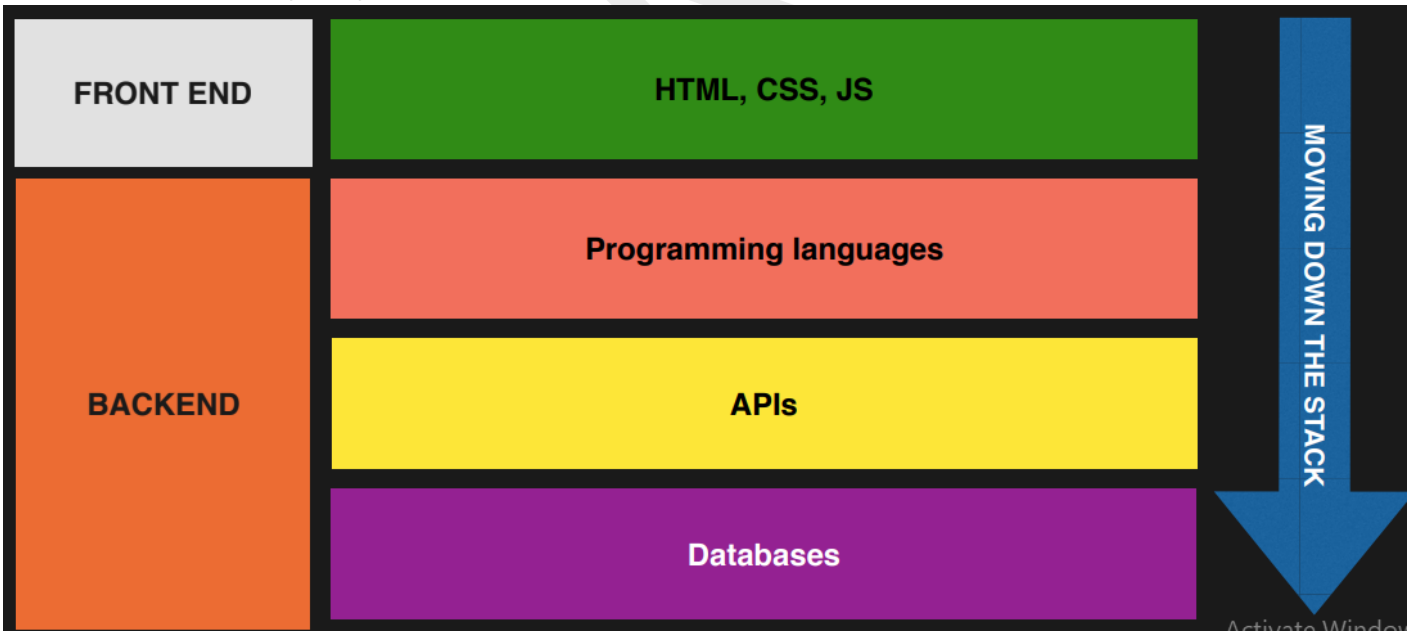
1. **Front - End / Client - side**
2. **Back - End / Server - side**

1. Front – End

- This is the part of the application that run in your browser.
- It's what the user sees and interacts with. It's the user interface (UI).

2. Back – End

- This is the part of the application that runs on a server.
- It's the "brain" that processes data, handles logins, talks to the database, and sends information back to the front-end to be displayed.



Part of Front-End

❖ **HTML**

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- HTML = Hyper Text Markup Language
- HTML is a markup language that tells the browser (chrome, firefox, safari, opera,.....) how to display content on a webpage(website). (HTML एक मार्कअप भाषा है जो ब्राउज़र (क्रोम, फ़ायरफ़ॉक्स, सफारी, ओपेरा,... ..) को बताती है कि वेबपेज (वेबसाइट) पर content कैसे प्रदर्शित की जाए।)
- It is the standard language used to create web pages.
- HTML uses tags to structure content such as headings, paragraphs, images, links, tables, etc.....
- Example : <h1>, <p>, <img>, <a>

❖ CSS

- CSS = Cascading Style Sheets
- It is used to style and design web pages.
- CSS is a style sheet language used to control the appearance of HTML elements on a webpage.(CSS एक style sheet भाषा हैं जो HTML के elements के दिखने के तरीके (design) को नियंत्रित करती हैं)
- With CSS you can change:- Colors, Fonts, Layout, Spacing, Background, Animation

❖ Javascript (JS)

- Javascript is a scripting language used to add behavior and interactivity to web pages. (java script एक scripting भाषा हैं जो webpage में behavior और interactivity जोड़ता हैं |)
- With javascript you can:- Show alerts and messages, Validate forms, Create animations, Update content without reloading page, Handle events (click, hover,key press).

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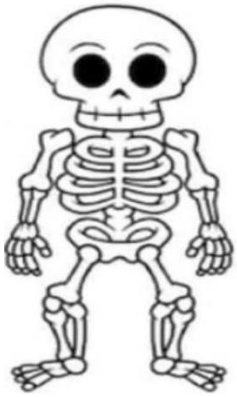


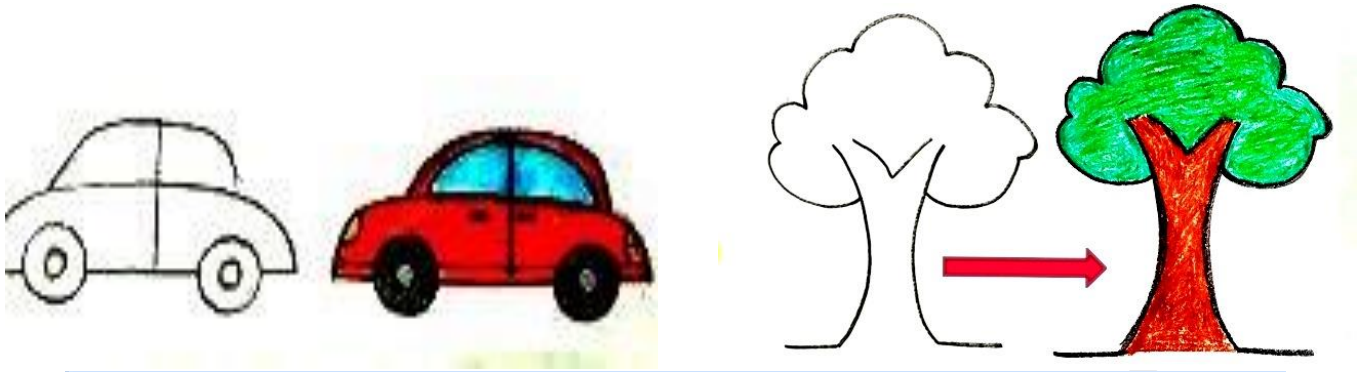
HTML

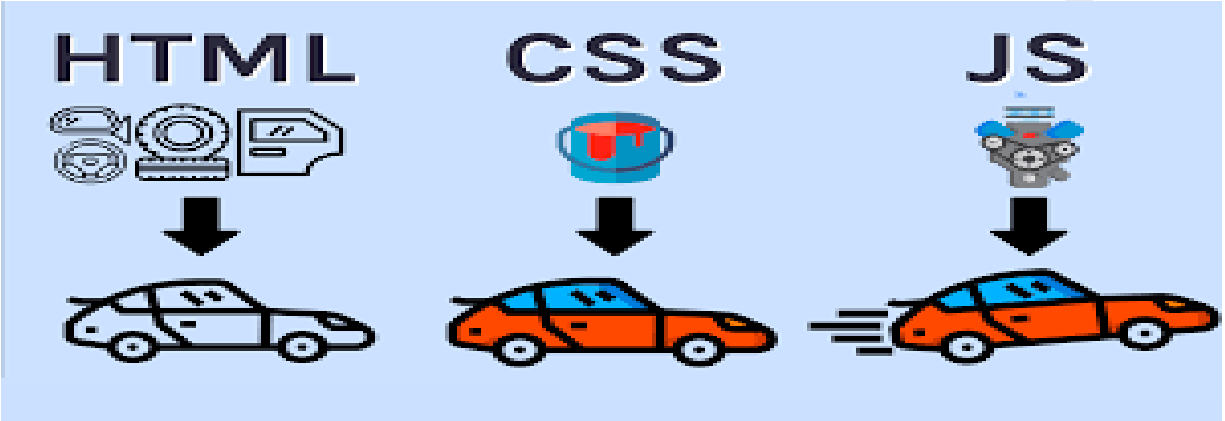
HTML + CSS

HTML+CSS+JavaScript

HTML

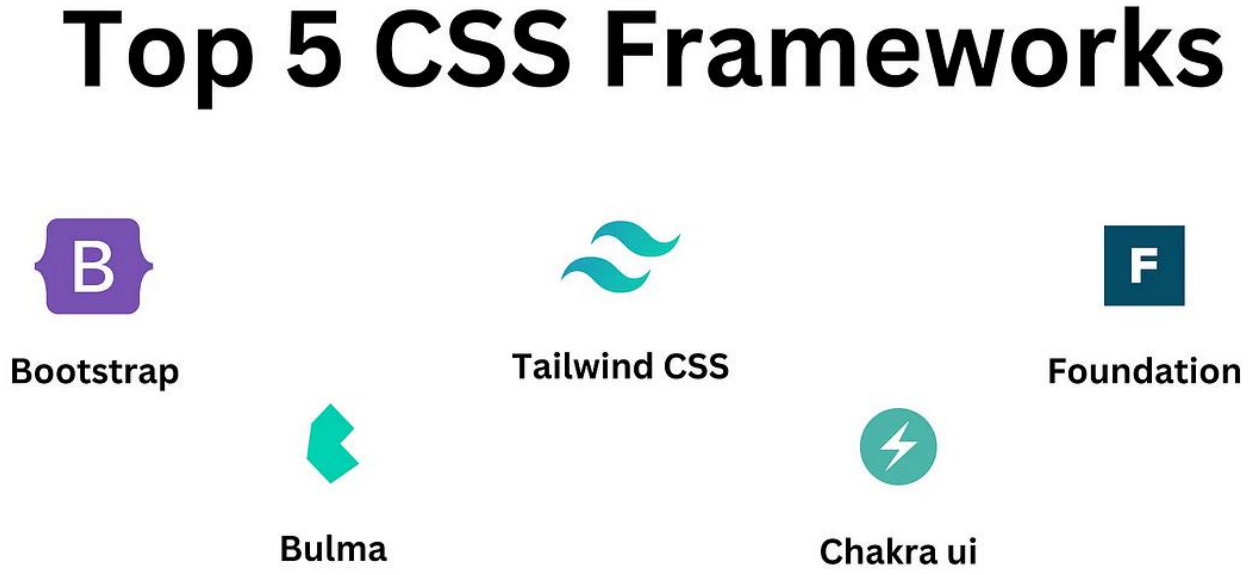
HTML + CSS





❖ What is Framework ?

- A framework is a pre-built structure or plateform that provides a foundation for developing software applications.
- It includes reusable code components, rules (step by step) , and tools that help developers build applications faster and more efficiently.
- Think of it as a ready – made template that saves you from writing everything from scratch.
- Use JS framework – React, Angular,Vue.js,….
- CSS frame work → Bootstrap, Tailwind CSS, Jquery (now, no demand),…..



- Example :-

Website	Framework
Facebook	React (react को fb ने ही बनाया)
Instagram	React
Netflix	React + Next.js
Airbnb	React
Twitter(X)	React

Whatsapp	React
Youtube	Polymer / Lit
Google (many product)	Angular
Github	React
Linkedin	React
Chess.com	Vue
Uber	React
Flipkart	React
Amazon	React (some part)
Spotify	React
BBC	Next.js
Alibaba	Vue.js
Xiaomi	Vue.js
Nike	React

High demand framework → React

❖ Without framework

1. Write everything from scratch/yourself.
2. More Bugs and Errors
3. Time Consuming
4. Difficult to scale
5. No standard structure
6. Security risks

❖ What is Library ?

- A library is a collection of pre-written code, functions, classes, and modules that you can use to perform common tasks without writing everything from scratch.

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OR

- A collection of resuable code that you can call upon when you need specific functionality.
- Think of it as a toolbox full of ready to use tools that make your development faster and eaiser.

❖ Difference b/w Framework and Library.

(कोई structure/template - book का , resume का , dress का,.....को mind में रख कर सोंचे)

1. Definition

✓ Library → It is a collection of pre-written functions or code that you can call when you need them.

✓ Framework → It is a complete structure/template that provides rules and flow for building an application. It controls how your code runs.

2. Control

✓ Library → You call the library.

✓ Framework → The framework calls your code.

3. Simple comparison table

Library	Framework
You control the flow	Framework controls the flow
Used for specific tasks	Provides full structure
Flexible	Structured with rules
You decide when to use it	It decides when your code runs

4. Real example

✓ React (mainlu for UI) → Library

✓ Angular → Full framework

✓ Vue.js → Progressive framework

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5. Simple real – life example

✓ Library → Tool Box
✓ Framework → House Blueprint

❖ React

- React is a JavaScript library for building user interfaces (UIs).
- It was created by facebook in 2013 (now Meta) and is currently one of the most popular front-end libraries in the world.
- Sometimes React is called a framework because it is often used with **Next.js**, which is built on top of React (react के ऊपर बना एक framework है) and provides full-stack features such as Server-Side Rendering (SSR), routing, and API routes.
- React → Library
- Next.js → React + Extra features

❖ Angular

- Angular is a comprehensive frontend framework developed and maintained by Google in 2016 (re-born) for building dynamic web applications.
- React (a type of library), where angular is a full – featured plateform that provides everything you need out of the box.

❖ Vue.js

- Vue.js is a javascript framework used to build user interfaces(UI) and single page applications (SPA).
- Created by Evan You (Ex- google employee) in 2014.
- Whose Use → Alibaba, Xiaomi, gitlab, Nintendo,….

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❖ Top framework in 2026

1. React & Next.js (most jobs and projects)
2. Vue.js (great for simpler and scalable apps)
3. Angular (Still strong in enterprise and large systems)
4. Svelte & svelteKit
5. Qwik
6. Astro (4,5,6→Growing fast for performance – focused sites)

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